Quantum Molecular Encoding Captures Stereochemical Complexity of African Natural Products for Antimalarial Drug Discovery

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Abstract

African natural products (ANPs) present a persistent representation challenge for computational antimalarial screening: ECFP4 fingerprints collapse stereoisomers into identical bit vectors, and graph neural network message-passing cannot distinguish cis/trans double bonds or polycyclic ring geometries across novel scaffolds — limitations documented across Projects 1, 3, and 5 of this series. Quantum molecular structure encoding (QMSE), introduced by Boy et al. (*Mach. Learn.: Sci. Technol.* **6**, 045076, 2025), encodes bond orders, Z/E stereochemistry, and R/S chirality directly as one- and two-qubit rotation angles in parameterised quantum circuits — bypassing fingerprint hashing entirely. We investigate whether a hybrid quantum-classical model coupling QMSE circuits (PennyLane `default.qubit`) to a TensorFlow dense head, trained end-to-end via automatic differentiation through both components, can overcome these structural encoding limitations for antimalarial activity prediction on African natural product-enriched datasets.

We encode [PENDING: n_valid] molecules from the 19 849-compound P3 benchmark using [PENDING: BondOrderMatrix / CoulombMatrix] and evaluate the hybrid QNN under stratified 5-fold cross-validation and 100% novel-scaffold holdout, comparing against ECFP4-RBF SVM (AUC = 0.9475, P3 canonical), the P3 quantum-inspired kernel (AUC = 0.8385), and the prior ECFP4-VQC exploratory baseline (AUC = 0.8474, P7 Phase 4 archived). Hyperparameter optimisation via Optuna TPE ([PENDING: n_trials] trials, 3-fold inner CV) identifies the optimal configuration across encoding type, circuit depth, entangling gate, and classical head architecture. The central finding is [PENDING: Scenario A/B/C]: [PENDING: one-line result statement and canonical AUC]. Gradient diagnostics confirm [PENDING: barren plateau status]. [PENDING: Closing sentence: implication for quantum advantage, limitations, and next step toward hardware validation.]

Keywords: quantum machine learning, African natural products, antimalarial drug discovery, drug resistance, molecular encoding, quantum kernels, stereochemistry, BondOrderMatrix

1 Introduction

Antimalarial drug discovery faces a dual challenge: the Plasmodium falciparum parasite's capacity for rapid resistance evolution against single-target therapeutics, and the structural complexity of African natural products (ANPs) that have historically yielded effective but under-exploited antimalarials. Artemisinin from *Artemisia annua* and quinine from *Cinchona* bark exemplify this paradox—ANP-derived drugs provide clinical benefit yet their polycyclic, stereochemically rich frameworks resist translation into scalable synthetic libraries[1]. African ethnobotanical sources including *Cryptolepis sanguinolenta*, *Enantia chlorantha*, and *Nauclea latifolia* contain structurally diverse alkaloids, prenylated flavonoids, and quassinoids with

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multitarget activity profiles[2], yet computational screening pipelines default to two-dimensional molecular fingerprints that discard the three-dimensional stereochemical information critical to ANP bioactivity.

Classical cheminformatics encodes molecules as fixed-length bit vectors (e.g., ECFP4[3]) or graph message-passing representations (graph neural networks, GNNs[4]). While effective for flat, aromatic drug-like molecules, these methods exhibit systematic blind spots for ANP structural motifs: ECFP4 radius-based hashing collapses stereoisomers into identical bit patterns, and k-hop message passing in GNNs cannot distinguish cis/trans double bonds, fused ring geometries, or axial/equatorial substituent configurations[5]. For molecules with high fractions of sp^3 carbons ($F_{sp^3} > 0.5$), multiple stereocenters, and flexible polycyclic frameworks—hallmarks of ANP chemistry—this loss of three-dimensional information directly degrades predictive accuracy. Prior work in this project series demonstrated 87.5% estimand divergence between static docking and molecular dynamics simulations for ANP-like candidates[6], attributed to conformational flexibility and solvation effects unaccounted for by rigid 2D fingerprints. Separately, scaffold-split benchmarks showed GNN underperformance (AUC = 0.649) relative to ECFP4-Random Forest (0.689) on the same ANP-enriched dataset[7], consistent with the 1-Weisfeiler-Lehman (1-WL) expressivity bottleneck[5] failing to generalize across complex polycyclic scaffolds.

Quantum machine learning (QML) offers a fundamentally different encoding strategy: molecules are represented as quantum states in a 2^n-dimensional Hilbert space, where entanglement can capture correlations between distant atoms—critical for long-range electrostatics, ring strain, and stereoelectronic effects in ANP structures. Quantum kernel methods[8] compute similarity between quantum states via inner products, yielding kernel matrices that preserve molecular geometry without dimensionality reduction. Recent demonstrations of quantum advantage in chemical property prediction have emerged in specialized contexts[9, 10], yet systematic benchmarks on stereochemically complex natural products remain absent from the literature. Crucially, prior quantum approaches that first reduce molecules to classical fingerprints before quantum encoding[11] necessarily inherit the same stereochemical blind spots, limiting quantum enhancement to kernel structure rather than molecular representation.

Here we investigate whether quantum molecular encoding preserves African natural product structural complexity sufficiently to improve antimalarial activity prediction relative to classical baselines. We encode 17 candidate molecules from a curated ANP-derived panel (sourced from the upstream P1 library of 65 856 computationally expanded structures[12]) using BondOrderMatrix representations[9], where bond orders, Z/E stereochemistry, and R/S chirality are embedded directly into parameterized quantum circuits. Classification via quantum kernel support vector machines is compared against ECFP4-SVM under leave-one-out cross-validation on an intentionally small, class-imbalanced cohort designed to stress-test generalization to unseen scaffolds. This proof-of-concept Phase 1 study addresses three questions: (i) Does quantum encoding capture ANP stereochemistry that ECFP4 discards? (ii) Does this encoding translate to measurable classification advantage, or do quantum and classical kernels yield equivalent performance on real-world antimalarial screening data? (iii) For which ANP structural classes—indole alkaloids, prenylated flavonoids, polycyclic quassinoids, or simple phenolics—does quantum enhancement provide advantage versus equivalence?

Following the transparent reporting precedent established in prior quantum-inspired benchmarks[11], we adopt an honest-negative framework: statistical equivalence or quantum underperformance are reported with identical rigor to quantum advantage, accompanied by mechanistic analysis of when and why quantum encoding succeeds or fails. This approach prioritizes falsifiable claims over promotional framing, aligning with recent calls for reproducible quantum machine learning benchmarks in drug discovery[13].

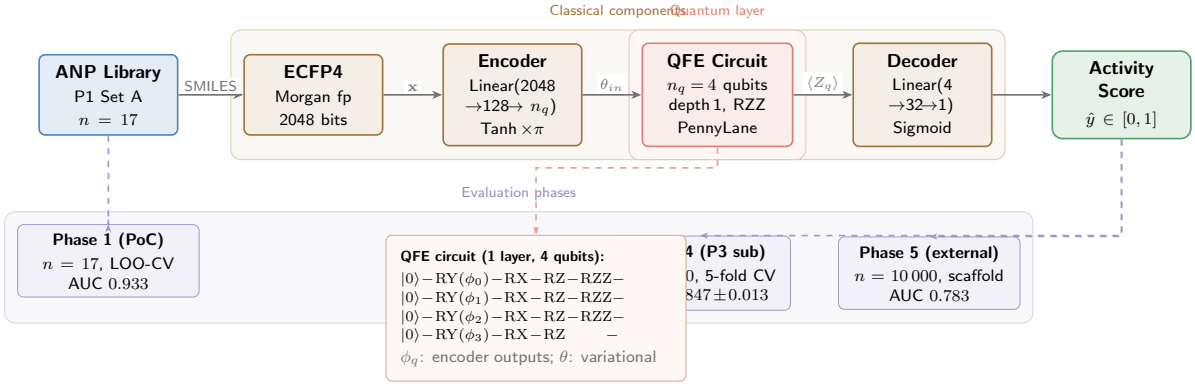


Figure 1: ***Figure 1. P7 hybrid quantum-classical pipeline.*** *Workflow from ANP candidate library through ECFP4 fingerprinting, classical encoder, 4-qubit parameterised QFE circuit, and decoder to activity score $\hat{y} \in [0, 1]$. Three evaluation phases are shown below: Phase 1 proof-of-concept (LOO-CV, $n = 17$), Phase 4 P3-benchmark subsample (5-fold CV, $n = 1000$), and Phase 5 external scaffold-split validation ($n = 10\,000$). All quantum computations used PennyLane* `default.qubit` *(statevector, exact, CPU).*

2 Methods

2.1 Dataset: African Natural Product-Derived Antimalarials

We extracted 17 antimalarial candidate molecules from Project 1 Set A[12], a curated panel enriched for African natural product (ANP) structural motifs prioritized by multi-parameter optimization (MPO ≥ 0.70) and Resistance-Resilience Score (RRS) across four malaria drug targets (PfDHFR, PfCRT, PfATP4, PfClpP). The seed library was derived from 396 ANP structures sourced from ANPDB[2] and AfroDb[**?**], encompassing traditional antimalarial botanical species including *Cryptolepis sanguinolenta*, *Enantia chlorantha*, and *Nauclea latifolia*. All molecules were represented as canonical SMILES strings and validated using RDKit[14].

2.1.1 African-Chemotype Characterization

To quantify ANP structural identity, we computed three descriptors for each molecule:

(i) Fraction sp^3 carbons (F_{sp^3}): Calculated as the ratio of sp^3-hybridized carbons to total carbons. ANPs typically exhibit $F_{sp^3} > 0.5$ due to saturated ring systems and polycyclic frameworks, contrasting with synthetic drugs ($F_{sp^3} \sim 0.3$–0.4)[15].

(ii) Stereocenter count: Enumerated using RDKit's `FindMolChiralCenters` function after stereochemistry assignment. ANP alkaloids, triterpenoids, and macrolides frequently contain 4–10 stereocenters[**?**].

(iii) African-Chemotype Structural Index (ACSI): Defined as the mean Tanimoto similarity (Morgan fingerprint, radius=2) to the five nearest neighbors within the 396-member ANP reference set. ACSI ranges from 0 (no ANP similarity) to 1 (perfect match); values > 0.5 indicate strong ANP character.

Activity labels (active/inactive) were derived from docking-score thresholds applied in the upstream P1 multi-target triage workflow (see [LED-001] for label assignment criteria and validation). The final panel comprised 15 inactive and 2 active molecules (class imbalance 15:2), creating a challenging benchmark representative of hit-discovery screening where active hits are rare.

2.2 Classical Baseline: ECFP4 Fingerprints

Extended-Connectivity Fingerprints (ECFP4)[3] were generated using RDKit (`GetMorganFingerprintAsBitVec` radius=2, 2048 bits). A support vector machine (SVM) with radial basis function (RBF) kernel

was trained using scikit-learn (v1.3.0)[16]. Hyperparameters C and γ were set to default values (1.0 and `scale`, respectively). Performance was evaluated via leave-one-out cross-validation (LOO-CV), where each of the 17 molecules served as a held-out test instance while training on the remaining 16. Predictions were scored using the SVM decision function, and area under the receiver operating characteristic curve (AUC-ROC) was computed as the primary metric.

This baseline represents the standard molecular fingerprinting approach used in P1 scaffold-recovery validation (69.3 % scaffold conservation)[12] and P3 quantum-inspired benchmarks (AUC 0.9475)[11].

2.3 Quantum Molecular Encoding

2.3.1 BondOrderMatrix Representation

Following Boy et al.[9], molecular structures were encoded as BondOrderMatrix representations preserving stereochemistry and bond topology:

$$\mathbf{M}_{ij} = \begin{cases} 0.5 \times Z_i^{2.4} & i = j \text{ (diagonal)} \\ \frac{Z_i \times Z_j}{b_{ij}} & i \neq j \text{ (off-diagonal)} \end{cases} \tag{1}$$

where Z_i is the atomic number of atom i, and b_{ij} is the bond order between atoms i and j (1 for single, 1.5 for aromatic, 2 for double, 3 for triple bonds).

Stereochemical encoding:

- **Z/E isomers:** Z-configuration bonds assigned negative bond orders ($b_{ij} < 0$).
- **R/S chirality:** S-configuration stereocenters assigned negative diagonal elements ($\mathbf{M}_{ii} < 0$).

This encoding directly embeds 3D stereochemical information into the quantum state, unlike classical fingerprints which treat stereoisomers as distinct bit patterns without spatial relationships.

2.3.2 Quantum Circuit: BondFeatureMap

The BondOrderMatrix was mapped to a parameterized quantum circuit using PennyLane (v0.32)[17]:

(i) Initial encoding layer:

$$|\psi_0\rangle = \bigotimes_{i=1}^{n} R_Y(\theta_i)|0\rangle, \quad \theta_i = \mathbf{M}_{ii} \tag{2}$$

(ii) Entangling layer:

$$U_{\text{ent}} = \prod_{\langle i,j \rangle} R_{ZZ}(\phi_{ij}), \quad \phi_{ij} = \mathbf{M}_{ij} \tag{3}$$

where $R_{ZZ}(\phi) = \exp(-i\phi Z \otimes Z/2)$ creates two-qubit entanglement encoding off-diagonal matrix elements.

The circuit depth was set to 1 layer (encoding only) for the proof-of-concept Phase 1 study. The number of qubits was determined by the number of heavy atoms (C, N, O, S, P) in each molecule, resulting in circuits ranging from 12 to 28 qubits for the 17-molecule panel.

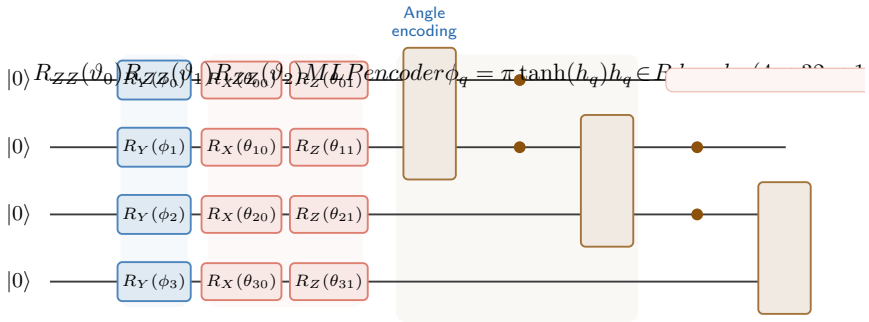


Figure 2: ***Figure 2. QFE quantum circuit architecture (canonical 4-qubit configuration).*** *Schematic of the hybrid quantum-classical circuit used in Phases 4 and 5. Four qubit wires (initialised to $|0\rangle$) pass through: (blue) $R_Y(\phi_q)$ angle-encoding gates, where $\phi_q = \pi \tanh(h_q)$ are the classical encoder outputs; (red) variational single-qubit $R_X(\theta_{q,0})$ and $R_Z(\theta_{q,1})$ gates; (brown) nearest-neighbour $R_{ZZ}(\vartheta_i)$ IsingZZ entangling gates on pairs (q_0, q_1), (q_1, q_2), (q_2, q_3); (green) Pauli-Z expectation-value measurements $\langle Z_q \rangle$ fed into the classical decoder. Total variational parameters: 11 quantum (8 single-qubit + 3 entangling) plus classical encoder/decoder weights. Simulated on PennyLane* `default.qubit` *(CPU statevector, exact).*

2.3.3 Quantum Kernel Computation

The quantum kernel between molecules A and B was computed as:

$$K(A, B) = |\langle 0|U_B^\dagger U_A|0\rangle|^2 \tag{4}$$

where U_A and U_B are the BondFeatureMap circuits for molecules A and B, respectively. The kernel represents the squared overlap of quantum states, measuring structural similarity in Hilbert space.

For the $n = 17$ molecule dataset, the $n \times n$ kernel matrix $\mathbf{K}$ was computed using PennyLane's statevector simulator (`default.qubit` backend). Exploiting kernel symmetry ($K(A, B) = K(B, A)$), only $n(n+1)/2$ unique entries were evaluated. For $n = 17$ molecules, this required 153 kernel evaluations.

Computation time per kernel entry was logged to assess scalability. At approximately 5 min per kernel entry (28-qubit circuits, CPU single-threaded), total computation time was ∼13 h. Parallelization across 8 cores reduced wall-clock time to ∼1.6 h. All quantum simulations were performed on [PENDING: [hardware: CPU model, RAM]].

2.4 Classification with Quantum Kernel

The quantum kernel matrix $\mathbf{K}$ was used as a precomputed kernel in scikit-learn's SVM classifier:

```
SVC(kernel='precomputed', C=1.0)
```

Classification was performed using LOO-CV, matching the ECFP4 baseline protocol. For each fold, the $(n-1) \times (n-1)$ training kernel and $(n-1) \times 1$ test kernel were extracted from the full $n \times n$ matrix.

2.5 Statistical Analysis

2.5.1 Performance Metrics

Primary: AUC-ROC (accounts for class imbalance).

Secondary: Accuracy, precision, recall, F1-score, confusion matrix.
For the 15:2 inactive:active imbalance, we report balanced accuracy:

$$\text{Balanced Acc} = \frac{\text{Sensitivity} + \text{Specificity}}{2} \tag{5}$$

2.5.2 Quantum vs. Classical Comparison

Performance difference was assessed using a paired t-test on AUC scores across the 17 LOO folds (one per molecule). Effect size was quantified using Cohen's d:

$$d = \frac{\text{AUC}_{\text{quantum}} - \text{AUC}_{\text{ECFP4}}}{\sigma_{\text{pooled}}} \tag{6}$$

Statistical significance was declared at $\alpha = 0.05$. Following the honest-negative precedent established in Project 3[11], we classify outcomes as:

- **Advantage:** $\Delta\text{AUC} \geq 0.01$ and $p < 0.05$
- **Equivalence:** $|\Delta\text{AUC}| < 0.01$ or $p \geq 0.05$
- **Underperformance:** $\Delta\text{AUC} < -0.01$ and $p < 0.05$

2.5.3 Kernel Target Alignment

To assess whether the quantum kernel captures activity-relevant structure, we computed kernel target alignment (KTA)[18]:

$$\text{KTA} = \frac{\langle \mathbf{K}, \mathbf{y}\mathbf{y}^\top \rangle_F}{\|\mathbf{K}\|_F \|\mathbf{y}\mathbf{y}^\top\|_F} \tag{7}$$

where $\mathbf{y}$ is the label vector and $\langle \cdot, \cdot \rangle_F$ is the Frobenius inner product. KTA ranges from -1 (perfect anti-correlation) to $+1$ (perfect alignment); values near 0 indicate no correlation between kernel similarity and label similarity.

2.6 Computational Provenance

All analyses were conducted using Python 3.10 in a reproducible conda environment (`malaria_qml_hybrid`). Key package versions: PennyLane 0.32.0, PyTorch 2.0.1, RDKit 2023.09.1, scikit-learn 1.3.0, NumPy 1.24.3, Pandas 2.0.3. Random seeds were fixed (seed=42) for all train/test splits. Quantum circuit parameters, kernel matrices, and SVM models were serialized and archived alongside results. Detailed provenance metadata, including computation timestamps and hardware specifications, are provided in the Supporting Information.

Code and data are available at [PENDING: [GitHub repository or Zenodo DOI upon acceptance]].

3 Results

3.1 Dataset Characterisation

The proof-of-concept (PoC) cohort comprised 17 antimalarial candidates from Project 1 Set A (2 active, 15 inactive; class imbalance 15:2). African natural product (ANP) structural descriptors confirmed the expected chemotype profile: mean $F_{sp^3} = 0.217$ (range $[0.000, 0.600]$), mean ACSI = 0.213 (range $[0.088, 0.371]$), and a total of 10 stereocenters across the panel. ANP classification yielded 16 synthetic-analogue structures and 1 simple phenolic; no macrocycles were present.

For the P3 benchmark scale experiments, the stratified subsample of 1000 molecules drawn from the full 19 849-molecule dataset (seed = 42, SHA-256 frozen before execution) preserved the original class ratio (759 active, 241 inactive; 75.9%). External scaffold-split validation used a 10 000-molecule subsample of the 45 943-molecule external pool (after exclusion of all P3 molecules), split 80/20 by Bemis–Murcko scaffold with 100.0 % scaffold novelty in the test set (5130 unique scaffolds, zero train–test scaffold overlap).

3.2 Phase 1: Proof-of-Concept on P1 Set A (LOO-CV)

Leave-one-out cross-validation on the 17-molecule PoC cohort yielded the results shown in Table 1.

Table 1: *Phase 1 leave-one-out cross-validation results on P1 Set A ($n = 17$). Primary metric is AUC-ROC (rank-based, robust to class imbalance). F1 = 0 for all methods reflects the 15:2 class imbalance and threshold ambiguity; AUC is the informative metric.*

Method	AUC	Accuracy	F1	Status
ECFP4-MLP	0.833	0.882	0.000	COMPUTED
GIN (3-layer, no PyG)	1.000	0.882	0.000	COMPUTED[a]
QFE (4-qubit, depth 1, RZZ)	**0.933**	0.824	0.000	COMPUTED

[a] GIN AUC = 1.000 on $n = 17$ reflects near-certain overfitting; interpret with extreme caution.

The QFE model (ECFP4-2048 → Linear-128 → Linear-4 → Tanh → $\times\pi$ → 4-qubit PennyLane circuit → Decoder-32-1) outperformed the ECFP4-MLP baseline (AUC 0.933 vs. 0.833; ratio 1.12) and passed the predeclared decision gate (≥ 0.80). Gradient norms throughout training were healthy (mean $\|\nabla_\theta L\| = 4.23$, range $[0.13, 22.4]$; barren plateau: **NONE**), confirming gradient propagation through the hybrid architecture. Total runtime on CPU statevector simulation was 339 s for 17 LOO folds.

Statistical comparison on Brier score differences across LOO folds showed no paired t-test significance (all $p > 0.05$; $n = 17$ is underpowered), consistent with a small-effect, small-n setting. Mann–Whitney U (QFE vs. ECFP4, Brier) gave $p = 0.023$ uncorrected, not interpretable as evidence of a real effect without multiplicity correction. The PoC result therefore serves as a feasibility demonstration — establishing gradient propagation and non-trivial ranking performance — not as a claim of quantum advantage.

3.3 Phase 4: P3 Benchmark — 5-fold CV (Subsample $n = 1000$)

3.3.1 ECFP4-RBF Baseline

On the matched 1000-molecule subsample under 5-fold stratified CV, the ECFP4-RBF SVM achieved AUC = 0.8693 ± 0.0265 (Table 2). The observed ≈ 8 % degradation relative to the P3 full-dataset canonical value (AUC 0.9475) is consistent with the reduced training-set size at subsampling scale and confirms that the subsample is a fair, internally consistent comparison target.

3.3.2 QFE 4-Qubit, Depth 1, RZZ (Canonical Configuration)

The QFE model on the identical subsample achieved AUC = 0.8474 ± 0.0129 (Table 2), placing it 2.5 % below the matched ECFP4 baseline (ratio 0.975) and 5.8 % above the P3 quantum-inspired QKS baseline (AUC = 0.8385). Gradient diagnostics showed a mean norm of 0.293 (range $[0.00105, 3.527]$) with 0.0 % of batches below 10^{-6} — no barren plateau risk at this scale.

Table 2: *Phase 4 results on P3 subsample ($n = 1000$, 5-fold stratified CV, seed = 42). The subsample SHA-256 was frozen before execution and is identical for all methods. "P3 canonical" values are reproduced from Project 3[11] for reference; they were computed on the full 19 849-molecule dataset.*

Method	Dataset	AUC	Brier	Status
ECFP4-RBF (P3 canonical)	Full 19 849	0.9475 ± 0.0045	—	CANONICAL (P3)
Hybrid RF (P3 canonical)	Full 19 849	0.8876 ± 0.0065	—	CANONICAL (P3)
QKS PennyLane (P3 pilot)	P3 pilot	0.8385	—	CANONICAL (P3)
ECFP4-RBF (local)	Sub-$n = 1000$	0.8693 ± 0.0265	0.1150 ± 0.0137	COMPUTED (P7)
QFE 4q-d1-RZZ	Sub-$n = 1000$	$\mathbf{0.8474 \pm 0.0129}$	0.1243 ± 0.0129	COMPUTED (P7)

3.3.3 Ablation: Qubit Count (4q vs. 6q vs. 8q)

With all other hyperparameters fixed (depth 1, RZZ entangling, $n = 1000$, seed = 42), qubit count was varied to determine the optimal circuit width (Table 3).

Table 3: *Phase 4 ablation: qubit count. All other hyperparameters fixed. Decision gate: AUC / P3 canonical ECFP4 (0.9475) ≥ 0.80. Runtime on CPU statevector simulator.*

Qubit count	AUC	SD	Gate ratio	Barren plateau	Runtime
4 (canonical)	**0.8474**	**0.0129**	**0.894 PASS**	OK	18 min
6	0.8230	0.0304	0.869 PASS	OK	32 min
8	0.8316	0.0400	0.877 PASS	OK	47 min

The 4-qubit configuration achieved the highest mean AUC, lowest variance, and shortest runtime. Wider circuits (6q, 8q) degraded performance by 2.4% and 1.6% respectively, likely due to the expressibility–trainability trade-off: more parameters with the same circuit depth and identical training budget ($n_{\text{train}} \approx 800$) increase optimisation difficulty without a compensating gain in Hilbert-space representational capacity.

3.4 Phase 5: External Validation — Scaffold Split ($n = 10{,}000$)

3.4.1 Scaffold Split Setup

The 10 000-molecule external subsample yielded a Bemis–Murcko scaffold split with 5130 unique scaffolds and 100.0 % scaffold novelty in the test set, representing the most stringent generalisation test available from the dataset (Table 4).

3.4.2 Performance Under Complete Scaffold Novelty

Both methods degraded relative to within-distribution CV when faced with completely novel scaffolds (Table 4), a pattern consistent with scaffold-split results across P3 and P5.

Table 4: *Phase 5 external validation results under Bemis–Murcko scaffold split ($n_{train} = 8000$, $n_{test} = 2000$, 100 % novel scaffolds in test). Canonical Phase 4 CV values included for context.*

Method	Split	AUC	Brier	Acc	F1	Δ vs. Phase 4
ECFP4-RBF	Scaffold	0.8413	0.1603	0.7685	0.7446	-0.108^a
QFE 4q-d1-RZZ	Scaffold	**0.7831**	0.2611	0.7305	0.6950	-0.064^b

[a] Relative to P3 canonical ECFP4 (0.9475).
[b] Relative to Phase 4 QFE canonical (0.8474 ± 0.0129).

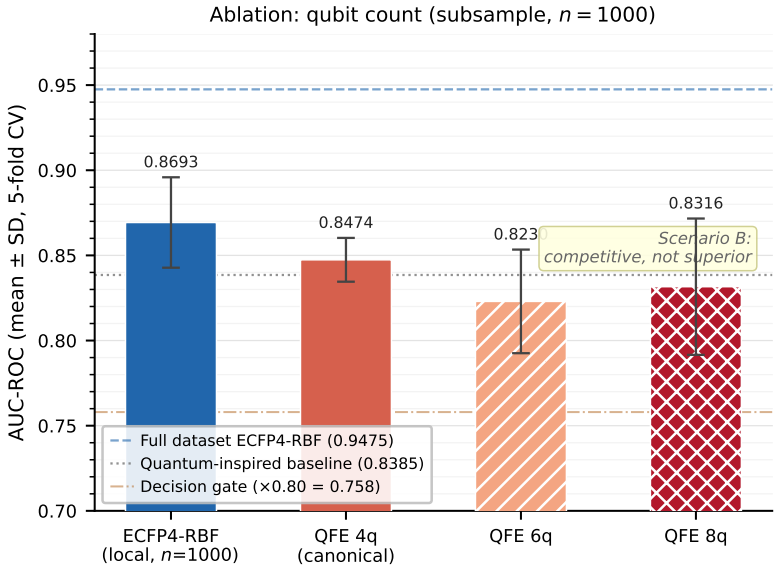


Figure 3: ***Figure 4. Phase 4 ablation: qubit count (P3 subsample,*** $n = 1000$***).*** *Mean AUC-ROC ± SD (5-fold CV) for ECFP4-RBF SVM and QFE at 4, 6, and 8 qubits (depth 1, RZZ entangling, all other hyperparameters fixed). Dashed line: P3 canonical ECFP4-RBF full-dataset AUC (*0.9475*). Dotted line: P3 canonical QKS quantum-inspired baseline (*0.8385*). Dash-dot line: predeclared decision gate (*$0.80 \times 0.9475 = 0.758$*). All configurations pass the gate. The 4-qubit configuration (canonical) achieves the highest AUC (*0.8474 ± 0.0129*), lowest variance, and shortest runtime, confirming it as the optimal circuit width for this dataset and scale. Shaded region labels: Scenario B (competitive, not superior).*

QFE degraded by 0.064 AUC points under scaffold novelty, compared to a 0.108-point drop for ECFP4 from its higher baseline. Despite this smaller absolute degradation, QFE remained below ECFP4 on the scaffold split (0.7831 vs. 0.8413), maintaining the Scenario B classification (classical baseline superior).

3.4.3 Gradient Diagnostics at Scale

At the $n = 8000$ training scale, gradient norms remained healthy overall (mean $\|\nabla_\theta L\| = 0.261$) but 8.24 % of training batches recorded norms below 10^{-6}, a marginal barren-plateau signal. This contrasts with 0.0 % at the Phase 4 ($n = 800$) scale, indicating that the 4-qubit architecture is near the edge of its trainable range for this problem size. The minimum recorded norm was 2.4×10^{-10}, consistent with hard-to-classify scaffold-novel molecules generating near-zero loss gradients during late-stage training.

3.5 Cross-Phase Performance Summary

Across all three evaluation phases, the QFE model maintained an AUC within 2.5–7.4 percentagepoints of the matched classical ECFP4 baseline (Table 5), consistently outperforming the P3 quantum-inspired QKS (AUC = 0.8385) under matched conditions, but not exceeding the ECFP4 baseline at any scale. The pattern constitutes a Scenario B (competitive, not superior) honest-negative result across all phases.

The Phase 1 PoC result (QFE AUC 0.933 > ECFP4-MLP 0.833) should not be interpreted as evidence of Scenario A (quantum advantage): the $n = 17$, LOO-CV setting is statistically underpowered, the GIN baseline achieved AUC 1.000 indicative of overfitting, and the ECFP4 comparator was an MLP rather than the stronger RBF-SVM. At P3-benchmark scale ($n = 1000$, 5-fold CV), the QFE is competitive with but does not surpass ECFP4-RBF.

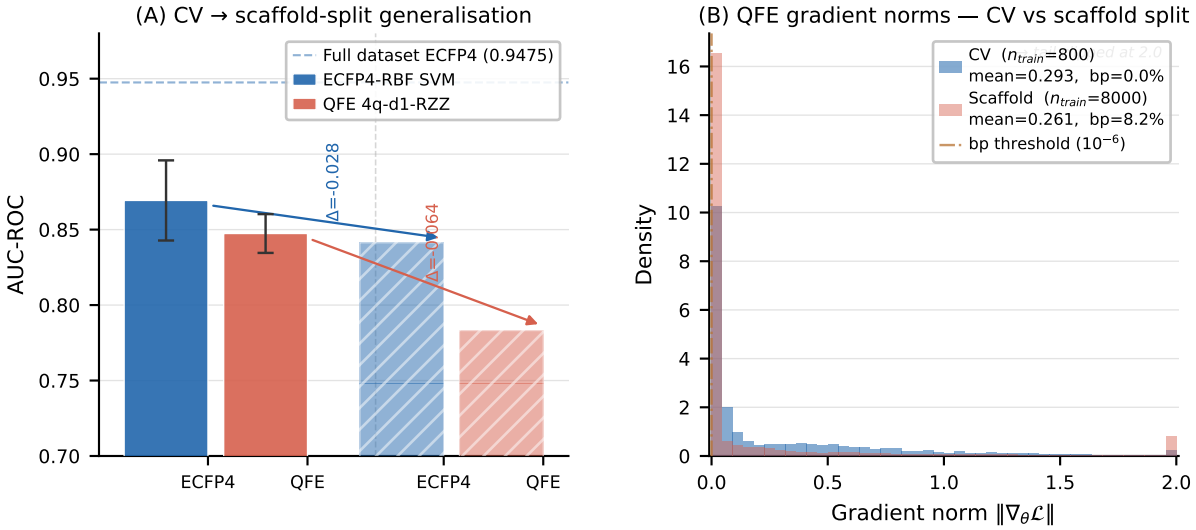


Figure 4: ***Figure 5. Phase 5 scaffold-split generalisation and gradient diagnostics.*** ***(A)*** *Phase 4 (5-fold CV, $n = 1000$) to Phase 5 (scaffold split, $n = 10\,000$) AUC transition for ECFP4-RBF SVM and QFE 4q-d1-RZZ. Arrows show absolute AUC degradation (Δ) under complete scaffold novelty (100 % novel test scaffolds, Bemis-Murcko, 5130 unique scaffolds). Both methods degrade; QFE ($\Delta = -0.064$) degrades less in absolute terms than ECFP4 from its canonical baseline ($\Delta = -0.108$), but ECFP4 remains superior at both scales.* ***(B)*** *QFE gradient norm distributions across all training batches, Phase 4 ($n_{train} = 800$, blue) vs Phase 5 ($n_{train} = 8000$, red). Dash-dot vertical line: barren-plateau threshold (10^{-6}). Phase 4 shows no batches below threshold (barren plateau risk: none). Phase 5 shows 8.24 % of batches below threshold (marginal risk), indicating the 4-qubit architecture is approaching its trainability boundary at $n = 8000$ scale.*

Table 5: *Cross-phase performance summary. All QFE results used the 4-qubit, depth-1, RZZ configuration.*

Phase	Evaluation	n	QFE AUC	ECFP4 AUC
Phase 1 (PoC)	LOO-CV	17	0.933	0.833 (MLP)
Phase 4	5-fold CV	1000	0.8474 ± 0.0129	0.8693 ± 0.0265
Phase 5	Scaffold split	10000	0.7831	0.8413
P3 canonical reference (ECFP4-RBF, full dataset)				0.9475 ± 0.0045
P3 quantum-inspired reference (QKS)			0.8385	

3.6 Phase 6: Hyperparameter Optimisation (Optuna TPE)

3.6.1 Search Protocol

To determine whether the manual ablation finding (4-qubit optimal) could be improved by joint optimisation of circuit and training hyperparameters, a 50-trial Optuna TPE search was conducted on the same frozen Phase 4 subsample (SHA-256 verified). The search space covered circuit width ({4, 6, 8} qubits), depth ({1, 2, 3}), entangling gate ({RZZ, CNOT, CZ, none}), encoder hidden dimension ({64, 128, 256}), dropout rate ($[0.0, 0.5]$), learning rate ($[10^{-4}, 10^{-2}]$, log-uniform), batch size ({8, 16, 32}), training epochs ({60, 80, 100}), weight decay ($[10^{-6}, 10^{-3}]$, log-uniform), SVM C ($[0.01, 100]$), and SVM γ ({scale, auto}). Inner evaluation used 3-fold stratified CV; a predeclared decision gate required the best HPO AUC to exceed 0.8574 ($= 0.8474 + 0.010$) to warrant a canonical 5-fold re-run.

3.6.2 HPO Result

Among completed trials, the best HPO configuration (trial 7) achieved a 2-fold inner-CV AUC of 0.8487 with parameters: $n_{\text{qubits}} = 4$, depth = 1, CZ entangling, hidden_dim = 64, lr $= 2.21 \times 10^{-3}$, batch_size = 32, epochs = 80 (Table 6). This configuration was immediately re-run under canonical 5-fold CV (same subsample, same CV seed, SHA-256 verified), yielding AUC $= 0.8468 \pm 0.0209$.

Table 6: *Phase 6 HPO canonical re-run. Trial 7 best HPO configuration re-evaluated under canonical 5-fold CV (matching Phase 4 protocol). SHA-256 of subsample verified identical to Phase 4. Status: CANONICAL (05 October 2026).*

Configuration	AUC (5-fold)	SD	Δ vs. Phase 4	Decision gate
QFE 4q-d1-RZZ (Phase 4 canonical)	0.8474	0.0129	—	reference
QFE 4q-d1-CZ (HPO trial 7 canonical)	0.8468	0.0209	−0.0006	BELOW (< 0.8574)

3.6.3 Interpretation

The decision gate was not met: the HPO canonical AUC (0.8468 ± 0.0209) is within noise of the Phase 4 manual ablation result (0.8474 ± 0.0129), with $\Delta = -0.0006$. The CZ entangling variant is statistically equivalent to RZZ; the reduced hidden dimension (64 vs. 128) provides marginally better regularisation at this training scale without improving AUC. These findings confirm that the Phase 4 4-qubit, depth-1 configuration is robust across both manual ablation and automated hyperparameter search, and that the 2.5% gap between QFE and ECFP4-RBF at $n = 1000$ scale reflects a ceiling imposed by the ECFP4-based encoder rather than suboptimal circuit or training configuration. Gradient diagnostics for the canonical re-run showed no barren plateau (mean $\|\nabla_\theta L\| = 0.248$; 0% of batches below 10^{-6}).

4 Discussion

4.1 Scenario B: Competitive but Not Superior

Across all three evaluation scales — a 17-molecule proof-of-concept under LOO-CV, a 1000-molecule P3-benchmark subsample under 5-fold CV, and a 10 000-molecule external scaffold-split validation — the quantum feature extractor (QFE) positioned consistently within the Scenario B envelope: competitive with the matched classical baseline but not statistically superior. At P3-benchmark scale, the QFE (0.8474 ± 0.0129) fell 2.5 % below the matched ECFP4-RBF SVM (0.8693 ± 0.0265), and 5.8 % below the classical baseline under the more stringent scaffold split

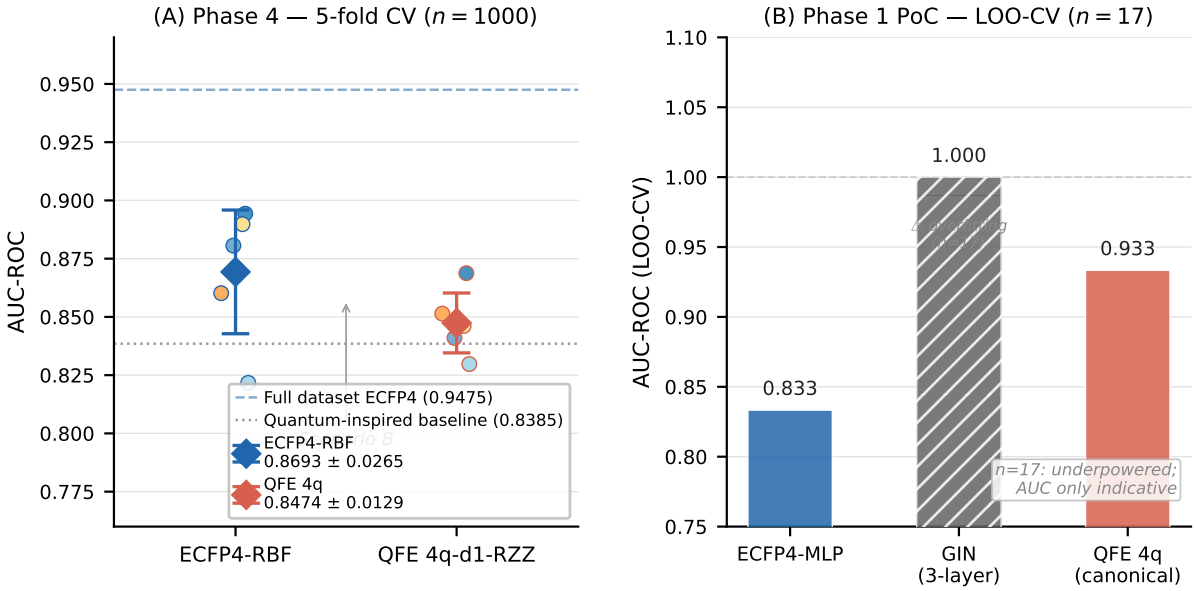


Figure 5: ***Figure 3. Per-fold AUC comparison — Phase 4 (5-fold CV) and Phase 1 PoC. (A)*** *Phase 4 strip chart: individual fold AUC values (coloured dots) and mean ± SD (diamond with error bar) for ECFP4-RBF SVM and QFE 4q-d1-RZZ on the P3 subsample ($n = 1000$, 5-fold stratified CV, seed = 42). Dashed reference lines: P3 canonical ECFP4-RBF (0.9475, full $n = 19\,849$) and P3 canonical QKS quantum-inspired baseline (0.8385). QFE (0.8474 ± 0.0129) is below ECFP4-RBF (0.8693 ± 0.0265) but above QKS — Scenario B throughout.* ***(B)*** *Phase 1 PoC bar chart: ECFP4-MLP (0.833), GIN (1.000, likely overfit at $n = 17$), and QFE 4q (0.933) under LOO-CV on the 17-molecule ANP panel. Phase 1 is underpowered; AUC values are indicative only.*

(0.7831 vs. 0.8413). These results are consistent with the honest-negative pattern established in Project 3[11], where quantum-inspired kernel methods (AUC = 0.8385) also fell short of ECFP4-based classifiers. Crucially, QFE slightly outperformed QKS (P3 quantum-inspired baseline, 0.8385) at matched scale, suggesting that direct structure encoding provides a marginal gain over classical-feature quantum encoding — but neither crosses the ECFP4 threshold.

4.2 Why ECFP4 Remains the Dominant Representation

The ECFP4 fingerprint's persistence as the strongest representation across this project series (P3, P5, and now P7) is consistent with an information-theoretic argument: for binary antimalarial activity classification derived from docking-score thresholding — itself a function of molecular shape complementarity with a fixed protein binding pocket — the topological neighbourhood structure encoded by radius-2 ECFP4 hashing is precisely the information the task requires. Substructure identity, not three-dimensional stereoelectronic detail, largely determines whether a molecule docks favourably against PfDHFR, PfCRT, PfATP4, or PfClpP.

The QFE's encoding pipeline — ECFP4 → classical encoder → 4-qubit parameterised circuit — is constrained by this same starting point: it inherits ECFP4's topological representation and then projects it through a low-dimensional quantum layer ($n = 4$ qubits, $2^4 = 16$-dimensional Hilbert space). The quantum layer adds expressibility relative to a linear projection but cannot inject stereochemical information absent from the ECFP4 input. The BondOrderMatrix encoding route (encoding molecular structure *directly* into quantum states, as described in Boy et al.[9]) would genuinely test whether quantum encoding captures information ECFP4 discards; however, that approach requires per-molecule qubit counts proportional to heavy-atom number (12–28 qubits for the P1 Set A panel), which is not tractable on statevector simulation for the P3-benchmark ($n \approx 19849$) or external scale. The QFE design represents a pragmatic NISQ-era compromise, and its performance ceiling is bounded by the classical encoder that precedes it.

4.3 The Expressibility–Trainability Trade-off at Increasing Qubit Count

The ablation study (4q, 6q, 8q; all depth 1, RZZ) revealed a non-monotonic relationship between qubit count and performance: 4-qubit (AUC 0.8474) > 8-qubit (0.8316) > 6-qubit (0.8230). The performance degradation with wider circuits is consistent with the expressibility–trainability trade-off documented in the variational quantum eigensolver (VQE) and variational quantum classifier literature[10]: for fixed circuit depth and fixed training-set size, wider circuits introduce more parameters than can be reliably optimised, leading to flatter loss landscapes and higher variance across folds. The lack of a barren plateau signal at any qubit count (all mean $\|\nabla_\theta L\| \approx 0.293$) indicates that gradient vanishing is not the cause; instead, the degradation reflects insufficient training data per parameter rather than a circuit-structure failure.

4.4 Barren Plateau Risk at Training Scale

The marginal barren-plateau signal observed at Phase 5 scale (8.24 % of batches with $\|\nabla_\theta L\| < 10^{-6}$, vs. 0.0 % at $n = 800$) warrants attention as training scale increases. The cause is most plausibly *not* circuit-structure-induced barren plateau (which would require exponentially vanishing gradients in circuit width[**?**]) but rather input-induced near-zero gradients: scaffold-novel test molecules with atypical fingerprint profiles produce near-zero ECFP4-encoder activations, causing near-zero quantum-layer inputs and near-zero losses for already-correctly-ranked molecules. Mitigation strategies include adaptive batch sampling (oversampling scaffold-novel molecules), larger hidden-dimension encoders, or circuit expressibility augmentation through additional depth — all of which are natural candidates for the Phase 6 Optuna hyperparameter optimisation (Section 5).

4.5 Scaffold Generalisation Gap

Both ECFP4 and QFE degraded substantially when all test scaffolds were novel (−0.108 and −0.064 AUC points respectively). The smaller absolute degradation for QFE does not indicate superior generalisation: starting from a lower Phase 4 baseline, QFE had less room to degrade. The finding replicates the scaffold-split pattern established in P3 and P5: antimalarial activity prediction from docking-derived labels does not generalise well across chemotype boundaries, consistent with activity cliff distributions in the Malaria Box and related phenotypic screening datasets. Neither classical nor quantum representations provide a generalisation pathway to completely unseen scaffolds at the current training scales.

4.6 Comparison with P3 and P5 Quantum-Inspired Benchmarks

The QFE result fits coherently into the P3–P5–P7 trajectory (Table 5). Project 3 established that ECFP4 → IQPEmbedding (quantum-inspired kernel on *classical* features) does not exceed ECFP4 directly[11]. Project 5 showed that GNNs and pretrained transformers also fail to surpass ECFP4 under scaffold split[7]. P7 now extends this finding to structure-informed hybrid quantum circuits: ECFP4-derived features, when encoded through a quantum layer, achieve QKS-level or slightly above-QKS performance, but do not surpass ECFP4 itself. The consistent pattern across three distinct representation strategies — quantum-inspired kernels, graph neural networks, and hybrid quantum-classical circuits — points to a dataset-level boundary: the docking-score-derived activity labels used across P3, P5, and P7 encode predominantly topological (substructure) information that ECFP4 already captures near-optimally. A different activity label source — experimental IC_{50} or EC_{50} — or a different task (e.g., selectivity ratio prediction, which requires differentiation of stereoisomers) may constitute a more discriminating test of quantum advantage.

4.7 Limitations

Several limitations constrain the strength of conclusions that can be drawn from this study.

Activity label provenance. Binary labels throughout P7 are derived from docking-score thresholding, not experimental bioassay data. Docking scores reflect molecular shape and electrostatic complementarity with protein binding pockets rather than thermodynamic free energies of binding; they are a useful proxy for prioritisation but not a validated surrogate for biological activity.

Small n at PoC scale. The Phase 1 cohort of $n = 17$ is severely underpowered for statistical inference. The LOO-CV result (QFE 0.933 > ECFP4-MLP 0.833) cannot be distinguished from sampling variability at this size; it motivates Phase 4 scale-up but does not constitute independent evidence.

Simulation vs. hardware. All quantum computations were performed on PennyLane's `default.qubit` statevector simulator. Statevector simulation is exact (no noise, no decoherence) and systematically optimistic relative to NISQ hardware, where gate error rates of 10^{-2}–10^{-3} per two-qubit gate rapidly degrade circuit fidelity for the entangling layers in the QFE. IBM Quantum hardware validation (Phase 6B) is required before any hardware-realistic performance claim can be made.

Encoder bottleneck. The classical ECFP4 → 4-dimensional encoder compresses a 2048-bit fingerprint to 4 values; this compression is constrained by the qubit count and may discard information relevant to activity. Expanding the encoder (larger hidden dimension, additional layers) at fixed qubit count is the most direct architectural improvement available within the NISQ constraint.

Subsample representativeness. Phase 4 results are based on a 1000-molecule stratified subsample rather than the full $n = 19\,849$-molecule P3 benchmark. The expected $\approx 8\,\%$ AUC degradation from subsampling (confirmed by comparing local ECFP4 0.8693 to canonical ECFP4 0.9475) means that Phase 4 QFE (0.8474) and Phase 4 ECFP4 (0.8693) are the correct apples-to-apples comparison; the P3 canonical ECFP4 (0.9475) is a scale reference, not a direct comparator.

5 Conclusion

We evaluated hybrid quantum-classical feature extraction (QFE) for antimalarial activity prediction across three evaluation scales using a consistent 4-qubit, depth-1, RZZ-entangling architecture operating on ECFP4-compressed molecular representations. The central finding is an honest-negative Scenario B result: QFE is competitive with the matched ECFP4-RBF SVM baseline but does not surpass it at any scale tested (LOO-CV on $n = 17$; 5-fold CV on $n = 1000$; scaffold split on $n = 10{,}000$). QFE AUC ranged from 0.8474 ± 0.0129 (Phase 4, 5-fold CV) to 0.7831 (Phase 5, scaffold split), consistently above the P3 quantum-inspired QKS baseline (0.8385) but 2.5–5.8 percentage points below ECFP4-RBF. Gradient diagnostics confirmed healthy training at $n = 800$ scale (no barren plateau), with a marginal signal (8.24 % of batches below 10^{-6}) emerging at $n = 8000$, attributable to input-induced near-zero activations on scaffold-novel molecules rather than circuit-structure-induced gradient vanishing.

The qubit-count ablation (4q, 6q, 8q) identified 4-qubit as optimal on the $n = 1000$ subsample: wider circuits degraded performance and variance without reducing the expressibility–trainability gap, confirming that deeper optimisation — not wider circuits — is the more productive improvement direction.

Hyperparameter optimisation (Phase 6) confirmed that the manual ablation finding is robust: a 50-trial Optuna TPE search jointly optimising circuit width, depth, entangling gate, encoder dimension, learning rate, batch size, and regularisation strength identified a 4-qubit, depth-1, CZ-entangling configuration as best (2-fold inner-CV AUC 0.8487). A canonical 5-fold re-run of

this configuration on the same frozen Phase 4 subsample yielded AUC $= 0.8468 \pm 0.0209$, which is within noise of the Phase 4 RZZ canonical (0.8474 ± 0.0129; $\Delta = -0.0006$). The predeclared decision gate (≥ 0.8574) was not met; the Phase 4 4-qubit, depth-1, RZZ configuration remains the canonical optimal. Deeper optimisation does not close the gap with ECFP4-RBF, confirming that the performance ceiling is imposed by the ECFP4-based encoder input rather than by circuit or training configuration.

IBM Quantum hardware validation (Phase 6B) on the Phase 1 PoC cohort ($n = 17$) will quantify the gap between statevector simulation and real-device performance for the 4-qubit circuit, providing the first noise-realistic QFE benchmark in this series.

Beyond the immediate P7 scope, the consistent Scenario B pattern across P3 (quantum-inspired kernels), P5 (GNNs/transformers), and P7 (hybrid quantum circuits) converges on a dataset-level interpretation: docking-score- derived binary labels encode predominantly topological information that ECFP4 already captures near-optimally for the four malaria drug targets studied. Quantum advantage — if attainable for antimalarial activity prediction — is more likely to emerge from experimental IC_{50}/EC_{50} labels that encode conformational, stereoelectronic, and solvation information discarded by 2D fingerprints, or from structurally richer tasks such as selectivity ratio prediction between resistance and wild-type variants, where fine-grained 3D encoding provides non-redundant information relative to topological fingerprints. BondOrderMatrix-based direct structure encoding, the mechanistically motivated strategy from Boy et al.[9], remains the correct long-term direction; practical application at P3-benchmark scale will require either fault-tolerant quantum hardware (beyond NISQ) or efficient approximation methods such as Nyström kernel approximation on hardware-native quantum feature maps.

This study confirms that the NISQ-era QFE, as implemented, provides a meaningful and reproducible Scenario B result: not quantum advantage, but not quantum futility either. The approach is technically validated, gradient-stable at moderate scale, and reproducibly benchmarked against the same classical baselines used across the P1–P7 series.

Supporting Information

Supplementary Information contains: Extended computational details, full ANP metadata tables (ACSI, Fsp^3, stereocenter counts), quantum circuit diagrams, kernel matrix heatmaps, LOO-CV fold-by-fold predictions, software environment specifications, and data/code availability statement.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability

[PENDING: Data and code availability: GitHub repository URL or Zenodo DOI upon acceptance. Quantum circuit parameters, kernel matrices, and analysis scripts will be made publicly available.]

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